

DARSINI LAKSHMIAH

darsinilakshmiah@gmail.com | (202)-766-3389 | [Linkedin](#) | [GITHUB](#) | [Portfolio](#)

Data Analyst with 4+ years of experience leveraging large-scale behavioral and transactional data to improve customer engagement, operational efficiency, and product decision-making across Amazon, AARP, and NatWest. Experienced in analytics engineering, experimentation, dashboarding, and ML-driven insights across retail, digital marketing, and banking environments.

WORK EXPERIENCE

AARP SERVICES INCORPORATED, Washington, DC, USA

Data Scientist Intern

September 2025 – March 2026

- Reduced manual campaign operations by **25%** by automating **30+** recurring marketing workflows previously executed in SAS using n8n, Databricks, PySpark, and Redshift, enabling scheduled execution, automated reporting, and campaign monitoring for analytics teams.
- Increased campaign engagement by **10%** by analyzing **76M+** user interaction records from S3 and Redshift and developing a LightGBM recommendation model to improve audience targeting across AARP digital content.
- Enabled marketing teams to monitor campaign performance in near real time by building Databricks and Redshift dashboards tracking CTR, open rates, engagement trends, and content-level KPIs.
- Reduced analytics turnaround time by **50%** by developing a Llama-powered analytics assistant that allowed marketing and product teams to query campaign performance and user behavior data from Databricks and Redshift using natural language.

AMAZON, Austin, Texas, USA

Business Intelligence Engineer Intern

May 2025 - August 2025

- Supported analytics for a **\$200M** multi-banner grocery retail initiative by developing a unified customer behavioral analytics platform integrating online and in-store purchase data across Amazon Fresh and Whole Foods, analyzing lifetime value and purchasing patterns across **87M+** transactions and **530+** store locations.
- Improved customer targeting and product experimentation capabilities by building a purchase-propensity prediction model using Python, SQL, and Amazon SageMaker; integrated model outputs into dashboards used by product and retail teams to evaluate feature launches and customer engagement trends.
- Reduced analytics latency and improved dashboard reliability by **20%** by engineering scalable ETL pipelines with AWS Glue, Athena, S3, and Redshift, transforming raw transactional data into optimized SQL aggregations powering retail analytics.
- Delivered operational insights for in-store digital features including Dash Cart and Click-Checkout by building performance dashboards and analyzing store-level adoption patterns; presented findings to stakeholders supporting the Amazon Fresh and Whole Foods integrated store strategy.

NATWEST GROUP, Chennai, India

Senior Data Analyst

Aug 2021 - June 2024

- Improved banking analytics query performance by **35%** by building distributed ETL pipelines and scalable data models using Azure Databricks, PySpark, and Azure Data Factory, processing **1.5M+** daily transactions across product and risk reporting systems.
- Strengthened AML and financial crime monitoring by developing an NLP-driven anomaly detection model using TF-IDF feature engineering and XGBoost classification on transaction data ingested into ADLS Gen2 and Snowflake, identifying suspicious behavioral patterns across UK and EMEA banking operations.
- Increased engagement for UK credit card and savings campaigns by **9%** through customer segmentation and A/B experimentation using Python and Snowflake; delivered performance insights and regulatory-compliant reporting through Power BI dashboards for business stakeholders.

EDUCATION

• **George Washington University, Washington, DC** - Master of Science in Data Science, GPA – 3.97/4.0

May 2026

• **Madras Christian College, Chennai, India** - Bachelor of Science in Mathematics, CGPA - 8.50/10

May 2021

PROJECTS

AI Interview Coach | [GitHub](#) | [Website](#) | AutoGen · GPT-4o · FastAPI · Whisper · WebSocket · face-api.js

- Built a real-time multi-agent interview simulation platform where AI interviewer, evaluator, and candidate agents conduct company-specific mock interviews over WebSocket with live Whisper transcription and webcam-based communication analysis for performance scoring.

Dock Wise AI Maritime Port Intelligence | [GitHub](#) | [Website](#) | ARIMA · Prophet · XGBoost · LangGraph · Groq · FAISS · Postgres

- Built a maritime supply chain intelligence platform analyzing live IMF Port Watch data across 117 US ports to forecast 14–28-day congestion trends; integrated a RAG-based AI assistant for natural-language port risk analysis and operational recommendations.

NutriNav Food Access Intelligence App | [GitHub](#) | [DevPost](#) | XGBoost · LLaMA 3.3 70B · Groq Whisper · FastAPI · Supabase

- Top 3 of 35 teams, George Hacks × UN Reboot the Earth. Built a multilingual voice-first app connecting 75,000+ DC SNAP residents to real-time corner store inventory; Whisper auto-detects language, LLaMA 3.3 70B responds in kind, and XGBoost grades products A - D via USDA Data

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL, SAS, PySpark
- **ML & AI:** Scikit-learn, TensorFlow, PyTorch, LangChain, LangGraph, AutoGen, LlamaIndex, Hugging Face
- **Cloud & Data Engineering:** AWS (Glue, Redshift, Athena, S3, Lambda, Bedrock), GCP BigQuery, Databricks, Snowflake, dbt, Airflow
- **Databases & Analytics:** MySQL, MongoDB, Apache Spark, Pandas, NumPy, A/B Testing, Causal Inference, KPI Measurement, Conversion Rate Analysis, Experimental Design.
- **Visualization & Tools:** Tableau, Power BI, QuickSight, Excel (Advanced), Git/GitHub, Docker, Flask.